

Positive integer exponents

For positive integer n , a to the n th power multiplies n copies of a .

$$a^n = a \cdot a \cdots a$$

What are the base and exponent in 7 to the fourth power?

Product rule

Use the same base.

$$a^m a^n = a^{m+n}$$

Why do we add the exponents?

A product of powers

$$x^4 x^7$$

How can we write this product as a single power?

Quotient rule

The base is nonzero.

$$\frac{a^m}{a^n} = a^{m-n}$$

What cancellation justifies this rule for positive integer exponents?

A quotient of powers

Assume x is nonzero.

$$\frac{x^9}{x^3}$$

How can we write this quotient as a single power?

Power of a power

$$(a^m)^n = a^{mn}$$

Why do we multiply the exponents here?

A power of a power

$$(x^3)^4$$

What single power is equivalent to this expression?

Zero exponents

The base is nonzero.

$$a^0 = 1$$

How does the quotient of two equal powers support this rule?

Negative exponents

The base is nonzero.

$$a^{-n} = \frac{1}{a^n}$$

What is the exact value of 3 to the power -2?

Negative exponents in a quotient

Assume x is nonzero.

$$\frac{x^2}{x^6}$$

How can we express the result using positive exponents?

Power of a product

$$(ab)^n = a^n b^n$$

How would we simplify the square of $3x$?

Power of a quotient

The denominator is nonzero.

$$\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$$

How would we simplify the cube of x divided by 2?

Combining exponent rules

Assume x and y are nonzero.

$$\frac{(2x^3y^{-2})^2}{4xy}$$

What equivalent expression uses only positive exponents?

A negative power of a fraction

Assume x and y are nonzero.

$$\left(\frac{2x}{3y}\right)^{-2}$$

What equivalent expression uses only positive exponents?

Different bases

$$2^3 \cdot 3^2$$

Can we combine these into a power by adding the exponents?

Powers of sums

$$(x + 2)^2$$

Does the power-of-a-product rule apply to this sum?

Scientific notation

For a positive number, scientific notation uses a coefficient from 1 up to, but not including, 10.

$$c \times 10^n, \quad 1 \leq c < 10$$

Why is 24 times 10 to the fifth power not normalized scientific notation?

A large number

6,800,000

How can we write this number in scientific notation?

A small number

0.000072

How can we write this number in scientific notation?

From scientific notation

$$4.2 \times 10^{-3}$$

What decimal does this expression represent?

Multiplication in scientific notation

$$(3 \times 10^5)(4 \times 10^{-2})$$

What is the product in normalized scientific notation?

Division in scientific notation

$$\frac{8.4 \times 10^7}{2.1 \times 10^3}$$

What is the quotient in scientific notation?